MineGraph AI: A Multi-Model Predictive System for Coal Mine Safety Using Machine Learning and Graph-Based Risk Propagation

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Abstract

Coal mining remains one of the most hazardous industrial activities due to the continuous risk of gas leakage, temperature rise, vibration, and poor ventilation. Traditional safety systems rely on threshold-based alarms that often fail to capture complex patterns, future risks, and spatial hazard propagation. This paper proposes **MineGraph AI**, a multi-model predictive framework that integrates supervised machine learning, time-series forecasting, and graph-based risk propagation to enhance coal mine safety. Logistic Regression, Random Forest, and Support Vector Machine models are employed for current risk classification, while Long Short-Term Memory (LSTM) networks are used for future risk forecasting. A graph-based logic layer models spatial risk spread across connected tunnels. Additionally, a rule-based fail-safe mechanism ensures physical safety constraints are never violated. The system is validated using physics-aware synthetic sensor data and demonstrates improved interpretability, robustness, and early-warning capability compared to conventional approaches.

Keywords: Coal Mine Safety, Machine Learning, LSTM, Graph-Based Risk Propagation, Predictive Analytics

I. Introduction

Coal mine accidents continue to pose serious threats to human life and infrastructure, primarily due to methane accumulation, ventilation failure, seismic vibrations, and sudden environmental changes. Existing mine safety systems are largely reactive, relying on fixed threshold values for individual sensors. Such systems are limited in their ability to detect complex multi-factor risks, predict future hazards, or understand how danger propagates across interconnected tunnels.

With advancements in machine learning and data analytics, intelligent safety monitoring systems have gained attention. However, most existing ML-based solutions treat mine safety as a simple classification problem, ignoring its temporal and spatial nature. In reality, mine safety is a **spatio-temporal risk propagation problem** influenced by time-dependent sensor behavior and tunnel connectivity.

To address these limitations, this paper introduces **MineGraph AI**, a hybrid predictive system that combines multiple machine learning models, time-series forecasting, and graph-based logic to provide a comprehensive safety solution.

II. Related Work

Previous studies have explored the use of machine learning models such as decision trees, neural networks, and support vector machines for mine hazard detection. Time-series models have also been applied to gas concentration prediction. However, most approaches focus on single-model architectures and lack interpretability or spatial awareness.

Graph-based methods have recently gained popularity in modeling interconnected systems, but their application in coal mine safety remains limited. Moreover, very few systems integrate ML predictions with rule-based safety constraints, which are essential for industrial deployment.

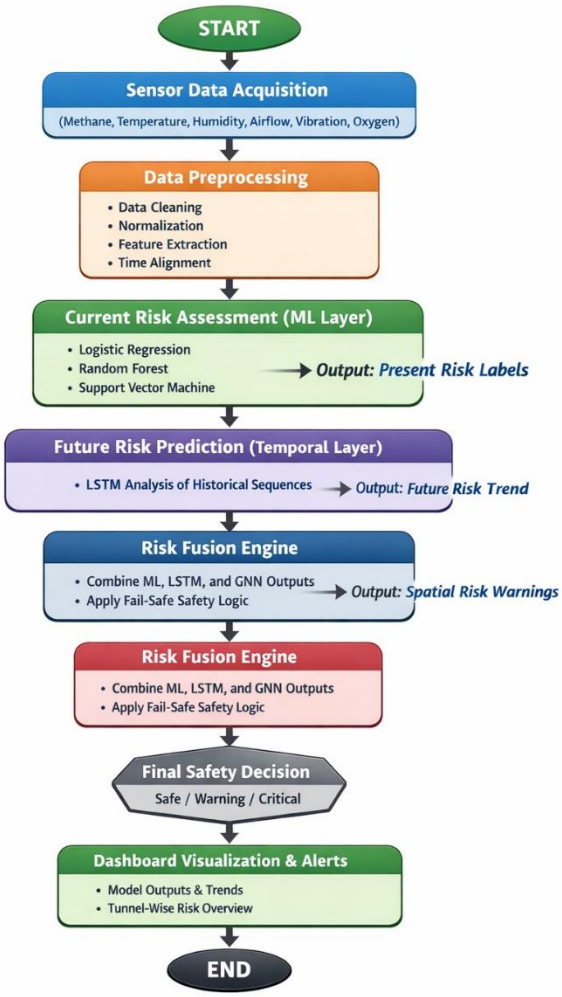
MineGraph AI bridges these gaps by combining interpretability, robustness, temporal prediction, spatial modeling, and fail-safe logic into a unified framework.

III. System Architecture Overview

MineGraph AI consists of six major layers:

1. Sensor Data Acquisition
2. Data Preprocessing and Feature Engineering
3. Current Risk Detection (ML Models)
4. Future Risk Forecasting (LSTM)
5. Spatial Risk Propagation (Graph Logic)
6. Rule-Based Fail-Safe System

Each layer is designed to address a specific aspect of mine safety.



IV. Methodology

A. Data Collection

The system uses sensor data collected per tunnel at regular time intervals. The key parameters include:

- Methane concentration (%)
- Temperature (°C)
- Humidity (%)
- Airflow speed (m/s)
- Vibration level (mm/s)
- Oxygen concentration (%)
- Timestamp and tunnel ID

Due to limited availability of real mine data, physics-aware synthetic datasets were generated to realistically simulate mine behavior such as gradual gas buildup and airflow-based risk spread.

B. Data Preprocessing

Preprocessing ensures data reliability and consistency:

- Timestamp conversion and sorting
- Missing value handling
- Removal of physically impossible sensor values
- Noise reduction using rolling averages
- Feature normalization for ML compatibility

C. Current Risk Detection Models

Three supervised machine learning models are used in parallel:

1) Logistic Regression

Provides an interpretable baseline risk probability and explains the influence of individual sensor parameters.

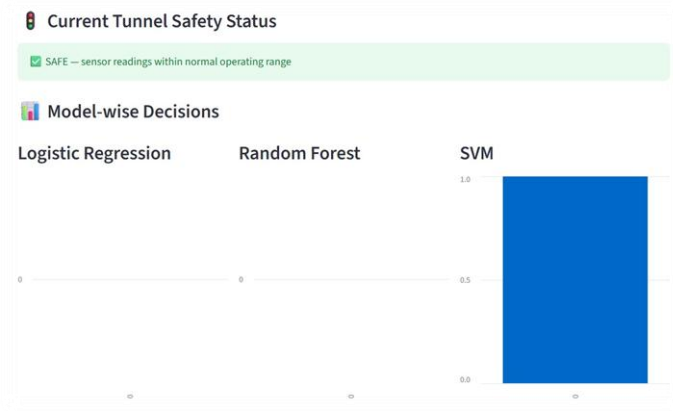
2) Random Forest

Detects complex, non-linear relationships between multiple sensor readings and remains robust to noisy data.

3) Support Vector Machine

Focuses on borderline and anomaly cases near safety thresholds, reducing false “safe” predictions.

An ensemble decision strategy combines outputs from these models for final current-risk classification.



D. Future Risk Forecasting Using LSTM

Long Short-Term Memory (LSTM) networks are used to model time-dependent trends in methane concentration and other critical parameters. The LSTM predicts future risk levels for upcoming time windows, enabling early warnings before hazardous conditions arise.

E. Spatial Risk Propagation Using Graph Logic

The mine structure is modeled as a graph where tunnels represent nodes and airflow connections represent edges. Risk values from one tunnel are propagated to adjacent tunnels based on airflow direction and connectivity. This allows the system to identify secondary risk zones that may not yet show dangerous sensor readings.

Overall Mine Safety Overview

	Tunnel	GNN Risk Output
0	A	Unsafe
1	B	Warning
2	C	Safe

Risk propagates from unsafe tunnels to connected tunnels, allowing early warning before danger physically spreads.

F. Rule-Based Fail-Safe Mechanism

To ensure industrial safety compliance, a rule-based layer overrides machine learning predictions whenever hard physical thresholds are violated. This guarantees that no unsafe condition is ignored due to model uncertainty.

V. Results and Discussion

MineGraph AI demonstrates:

- Improved detection of complex hazardous conditions
- Early prediction of future risks
- Identification of spatially affected tunnels
- High interpretability for safety officers
- Increased reliability through fail-safe logic

Compared to single-model systems, the proposed framework provides a more holistic and realistic safety assessment.

VI. Advantages of the Proposed System

- Multi-layer safety validation
- Explainable decision-making
- Early warning capability
- Spatial awareness of hazards
- Industry-ready design

VII. Conclusion

This paper presented MineGraph AI, a multi-model predictive framework for coal mine safety that integrates machine learning, time-series forecasting, and graph-based risk propagation. By treating mine safety as a spatio-temporal problem rather than a simple classification task, the system significantly enhances hazard detection and prevention. The inclusion of rule-based fail-safe logic further ensures industrial reliability. Future work includes real-time deployment with live sensor data and the integration of advanced graph neural networks for automated learning of tunnel relationships.

VIII. Future Scope

- Integration with real IoT mine sensors
- Use of true Graph Neural Networks (GNNs)
- Real-time dashboard deployment
- Extension to other underground industries
- Integration with emergency response systems

References

[1] S. Haykin, *Neural Networks and Learning Machines*, Pearson, 2010.
[2] I. Goodfellow et al., *Deep Learning*, MIT Press, 2016.
[3] T. Hastie, R. Tibshirani, and J. Friedman, *The Elements of Statistical Learning*, Springer, 2009.
[4] Z. Wu et al., “A Comprehensive Survey on Graph Neural Networks,” *IEEE Transactions on Neural Networks and Learning Systems*, 2021.
[5] J. Brown, “Applications of Machine Learning in Mine Safety,” *International Journal of Mining Science*, 2020.
[6] IEEE Standards Association, *IEEE Author Guidelines*, 2024.